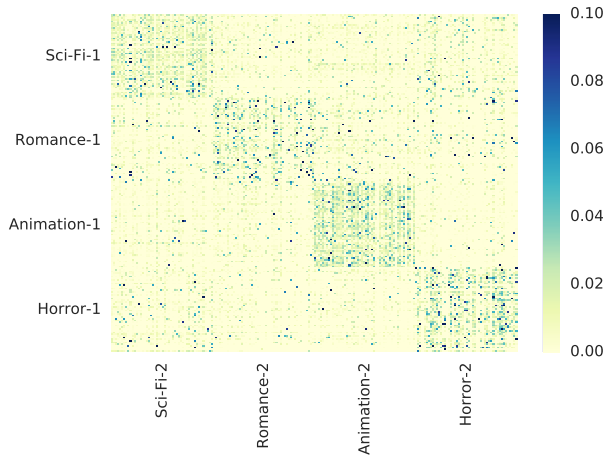

Figure 5: Visualization of average attention between movies from four categories. This shows our model can uncover items' attributes, and assigns larger weights between similar items.

REFERENCES

[1] S. Rendle, C. Freudenthaler, and L. Schmidt-Thieme, "Factorizing personalized markov chains for next-basket recommendation," in *WWW*, 2010.

[2] B. Hidasi, A. Karatzoglou, L. Baltrunas, and D. Tikk, "Session-based recommendations with recurrent neural networks," in *ICLR*, 2016.

[3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *NIPS*, 2017.

[4] Y. Hu, Y. Koren, and C. Volinsky, "Collaborative filtering for implicit feedback datasets," in *ICDM*, 2008.

[5] S. Rendle, C. Freudenthaler, Z. Gantner, and L. Schmidt-Thieme, "BPR: bayesian personalized ranking from implicit feedback," in *UAI*, 2009.

[6] F. Ricci, L. Rokach, B. Shapira, and P. Kantor, *Recommender systems handbook*. Springer US, 2011.

[7] Y. Koren and R. Bell, "Advances in collaborative filtering," in *Recommender Systems Handbook*. Springer, 2011.

[8] S. Kabbur, X. Ning, and G. Karypis, "Fism: factored item similarity models for top-n recommender systems," in *SIGKDD*, 2013.

[9] S. Zhang, L. Yao, and A. Sun, "Deep learning based recommender system: A survey and new perspectives," *arXiv*, vol. abs/1707.07435, 2017.

[10] S. Wang, Y. Wang, J. Tang, K. Shu, S. Ranganath, and H. Liu, "What your images reveal: Exploiting visual contents for point-of-interest recommendation," in *WWW*, 2017.

[11] W. Kang, C. Fang, Z. Wang, and J. McAuley, "Visually-aware fashion recommendation and design with generative image models," in *ICDM*, 2017.

[12] H. Wang, N. Wang, and D. Yeung, "Collaborative deep learning for recommender systems," in *SIGKDD*, 2015.

[13] D. H. Kim, C. Park, J. Oh, S. Lee, and H. Yu, "Convolutional matrix factorization for document context-aware recommendation," in *RecSys*, 2016.

[14] X. He, L. Liao, H. Zhang, L. Nie, X. Hu, and T. Chua, "Neural collaborative filtering," in *WWW*, 2017.

[15] S. Sedhain, A. K. Menon, S. Sanner, and L. Xie, "Autorec: Autoencoders meet collaborative filtering," in *WWW*, 2015.

[16] Y. Koren, "Collaborative filtering with temporal dynamics," *Communications of the ACM*, 2010.

[17] C. Wu, A. Ahmed, A. Beutel, A. J. Smola, and H. Jing, "Recurrent recommender networks," in *WSDM*, 2017.

[18] L. Xiong, X. Chen, T.-K. Huang, J. Schneider, and J. G. Carbonell, "Temporal collaborative filtering with bayesian probabilistic tensor factorization," in *SDM*, 2010.

[19] R. He, W. Kang, and J. McAuley, "Translation-based recommendation," in *RecSys*, 2017.

[20] R. He, C. Fang, Z. Wang, and J. McAuley, "Vista: A visually, socially, and temporally-aware model for artistic recommendation," in *RecSys*, 2016.

[21] R. He and J. McAuley, "Fusing similarity models with markov chains for sparse sequential recommendation," in *ICDM*, 2016.

[22] J. Tang and K. Wang, "Personalized top-n sequential recommendation via convolutional sequence embedding," in *WSDM*, 2018.

[23] H. Jing and A. J. Smola, "Neural survival recommender," in *WSDM*, 2017.

[24] Q. Liu, S. Wu, D. Wang, Z. Li, and L. Wang, "Context-aware sequential recommendation," in *ICDM*, 2016.

[25] A. Beutel, P. Covington, S. Jain, C. Xu, J. Li, V. Gatto, and E. H. Chi, "Latent cross: Making use of context in recurrent recommender systems," in *WSDM*, 2018.

[26] B. Hidasi and A. Karatzoglou, "Recurrent neural networks with top-k gains for session-based recommendations," *CoRR*, vol. abs/1706.03847, 2017.

[27] K. Xu, J. Ba, R. Kiros, K. Cho, A. C. Courville, R. Salakhutdinov, R. S. Zemel, and Y. Bengio, "Show, attend and tell: Neural image caption generation with visual attention," in *ICML*, 2015.

[28] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," in *ICLR*, 2015.

[29] J. Chen, H. Zhang, X. He, L. Nie, W. Liu, and T. Chua, "Attentive collaborative filtering: Multimedia recommendation with item- and component-level attention," in *SIGIR*, 2017.

[30] J. Xiao, H. Ye, X. He, H. Zhang, F. Wu, and T. Chua, "Attentional factorization machines: Learning the weight of feature interactions via attention networks," in *IJCAI*, 2017.

[31] S. Wang, L. Hu, L. Cao, X. Huang, D. Lian, and W. Liu, "Attention-based transactional context embedding for next-item recommendation," in *AAAI*, 2018.

[32] Y. Wu, M. Schuster, Z. Chen, Q. V. Le, M. Norouzi, W. Macherey, M. Krikun, Y. Cao, Q. Gao, K. Macherey *et al.*, "Google's neural machine translation system: Bridging the gap between human and machine translation," *arXiv preprint arXiv:1609.08144*, 2016.

[33] J. Zhou, Y. Cao, X. Wang, P. Li, and W. Xu, "Deep recurrent models with fast-forward connections for neural machine translation," *TACL*, 2016.

[34] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *ECCV*, 2014.

[35] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *CVPR*, 2016.

[36] L. J. Ba, R. Kiros, and G. E. Hinton, "Layer normalization," *CoRR*, vol. abs/1607.06450, 2016.

[37] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *ICML*, 2015.

[38] N. Srivastava, G. E. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: a simple way to prevent neural networks from overfitting," *JMLR*, 2014.

[39] D. Warde-Farley, I. J. Goodfellow, A. C. Courville, and Y. Bengio, "An empirical analysis of dropout in piecewise linear networks," *CoRR*, vol. abs/1312.6197, 2013.

[40] Y. Koren, R. Bell, and C. Volinsky, "Matrix factorization techniques for recommender systems," *Computer*, 2009.

[41] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *ICLR*, 2015.

[42] D. Povey, H. Hadian, P. Ghahremani, K. Li, and S. Khudanpur, "A time-restricted self-attention layer for asr," in *ICASSP*, 2018.

[43] P. Wang, J. Guo, Y. Lan, J. Xu, S. Wan, and X. Cheng, "Learning hierarchical representation model for next basket recommendation," in *SIGIR*, 2015.

[44] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *CoRR*, vol. abs/1803.01271, 2018.

[45] S. Hochreiter, Y. Bengio, P. Frasconi, J. Schmidhuber *et al.*, "Gradient flow in recurrent nets: the difficulty of learning long-term dependencies," 2001.

[46] J. J. McAuley, C. Targett, Q. Shi, and A. van den Hengel, "Image-based recommendations on styles and substitutes," in *SIGIR*, 2015.

[47] S. Feng, X. Li, Y. Zeng, G. Cong, Y. M. Chee, and Q. Yuan, "Personalized ranking metric embedding for next new poi recommendation," in *IJCAI*, 2015.

[48] Y. Koren, "Factorization meets the neighborhood: a multifaceted collaborative filtering model," in *SIGKDD*, 2008.